

Spotify 2023 Data Analysis using Power BI

Brief

You have been provided with the **Spotify 2023 dataset** ([spotify-2023.csv](#)), containing information about songs, artists, streaming performance, platform presence, playlists, charts, release information, and various audio characteristics.

For this coursework, consider yourself an **analyst working in the music industry**. Your task is to use **Microsoft Power BI** to explore the dataset, identify meaningful patterns and relationships, and develop insights that could support decision-making in the music industry.

You are expected to clean and prepare the data where necessary, create appropriate measures and calculated fields, develop meaningful visualizations, and present your findings through an **interactive Power BI report/dashboard**.

Analysis Tasks

1. Music Analysis

Explore patterns in the audio features of popular songs to understand musical characteristics and preferences.

Analyse variables such as **danceability, energy, valence, acousticness, instrumentality, liveness, speechiness, and BPM**. Identify important patterns and relationships between these characteristics and explain what they reveal about the musical characteristics of songs in the dataset.

Use suitable Power BI visualizations to support your analysis.

2. Cross-Platform Presence

Investigate how songs perform across different music platforms.

Analyse the presence and performance of songs across **Spotify, Apple Music, Deezer, and Shazam**, using available playlist and chart-related variables.

Identify songs that demonstrate strong presence across multiple platforms and investigate differences in platform performance. Present the findings using appropriate comparative Power BI visualizations.

3. Playlist Exposure and Streaming Performance

Investigate the relationship between a song's **playlist exposure and its streaming performance**.

Analyse the number of playlists associated with songs on the available platforms and compare this with their Spotify streams.

Identify whether songs with greater playlist exposure tend to have higher streaming numbers. Highlight significant patterns and unusual observations using Power BI.

4. Chart Position and Song Performance

Analyse the relationship between a song's **chart performance and its streaming performance**.

Use available Spotify, Apple Music, Deezer, and Shazam chart information to investigate whether songs with stronger chart positions also demonstrate higher streaming performance.

Create appropriate rankings, comparisons, and visualizations to identify patterns in chart performance and streams.

5. Outlier and Exceptional Song Analysis

Identify songs that behave differently from the general patterns in the dataset.

Use Power BI to identify **high-streaming songs with relatively low playlist or chart presence**, as well as songs with strong platform presence but comparatively lower streams.

Investigate the characteristics of these unusual songs and explain possible patterns using evidence from the dataset. Clearly distinguish between observations supported by the data and possible interpretations.

6. Artist Concentration and 80/20 Analysis

Investigate how streaming performance is distributed among artists.

Analyse whether a relatively small number of artists account for a large proportion of total streams in the dataset. Use appropriate Power BI techniques such as **cumulative totals, ranking, Pareto-style analysis, and percentage contribution**.

Identify the distribution of streaming performance across artists and discuss its implications for the music industry.

Instructions

1. The entire analysis must be conducted using **Microsoft Power BI**.
2. Students should import and appropriately prepare the supplied `spotify-2023.csv` dataset.
3. Appropriate **data cleaning, transformation, calculated columns, and DAX measures** should be used where required.
4. Students must select and justify suitable Power BI visualizations for each analysis.
5. The report should include appropriate **KPIs, slicers, filters, tables, charts, cards, and interactive elements**.
6. Students should provide meaningful interpretations rather than simply displaying charts.
7. Findings must be supported by evidence from the dataset.
8. Where relationships are identified, students should distinguish between **association/correlation and causation**.
9. Students should explain any assumptions made during the analysis.
10. The final Power BI report should be organized logically and provide an effective interactive user experience.
11. **Forecasting, predictive modelling, and future-value prediction are not required for this coursework.**
12. Students should provide a brief explanation of the methodology and rationale behind the analytical techniques and visualizations used.